

Lecture Notes in Computer Science

Edited by G. Goos and J. Hartmanis

372

G. Ausiello M. Dezani-Ciancaglini
S. Ronchi Della Rocca (Eds.)



Automata, Languages and Programming

16th International Colloquium
Stresa, Italy, July 11–15, 1989
Proceedings



Springer-Verlag

Berlin Heidelberg New York London Paris Tokyo Hong Kong

Editorial Board

D. Barstow W. Brauer P. Brinch Hansen D. Gries D. Luckham
C. Moler A. Pnueli G. Seegmüller J. Stoer N. Wirth

Volume Editors

Giorgio Ausiello
Università di Roma "La Sapienza"
Dipartimento di Informatica e Sistemistica
Via Buonarroti 12, I-00100 Roma, Italy

Mariangiola Dezani-Ciancaglini
Simonetta Ronchi Della Rocca
Università degli Studi di Torino
Dipartimento di Informatica
Corso Svizzera 185, I-10149 Torino, Italy

CR Subject Classification (1987): C.1, E.1, F, G.2, I.1, I.2.3

ISBN 3-540-51371-X Springer-Verlag Berlin Heidelberg New York
ISBN 0-387-51371-X Springer-Verlag New York Berlin Heidelberg

This work is subject to copyright. All rights are reserved, whether the whole or part of the material is concerned, specifically the rights of translation, reprinting, re-use of illustrations, recitation, broadcasting, reproduction on microfilms or in other ways, and storage in data banks. Duplication of this publication or parts thereof is only permitted under the provisions of the German Copyright Law of September 9, 1965, in its version of June 24, 1985, and a copyright fee must always be paid. Violations fall under the prosecution act of the German Copyright Law.

© Springer-Verlag Berlin Heidelberg 1989
Printed in Germany

Printing and binding: Druckhaus Beltz, Hemsbach/Bergstr.
2145/3140-543210 – Printed on acid-free paper

FOREWORD

ICALP is the Annual European Summer Conference on Theoretical Computer Science sponsored by the European Association for Theoretical Computer Science (EATCS).

ICALP stands for International Colloquium on Automata, Languages and Programming, but this conference series intends to cover all important areas of theoretical computer science such as: computability, automata theory, formal language theory, analysis of algorithms, computational complexity, mathematical aspects of programming language definition, logic and semantics of programming languages, foundation of logic programming, theorem proving, software specification, computational geometry, data types and data structures, theory of data bases and knowledge based systems, cryptography, VLSI structures, parallel and distributed computing, models of concurrency and robotics.

Previous colloquia were held in Tampere (1988), Karlsruhe (1987), Rennes (1986), Nauplion (1985), Antwerp (1984), Barcelona (1983), Aarhus (1982), Haifa (1981), Amsterdam (1980), Graz (1979), Udine (1978), Turku (1977), Edinburgh (1976), Saarbrücken (1974) and Paris (1972). ICALP 90 will be held in Great Britain, from July 16 to July 20, 1990, in Warwick.

ICALP 89 has been organized by the University of Torino (Italy). The conference is held in a very attractive location; Stresa, a small town on Lago Maggiore.

188 extended abstracts and draft papers were submitted to ICALP 89. The time schedule of the conference allowed for the acceptance of only 45 of these. Therefore the selection was very difficult and many good papers had to be rejected. Every paper was sent to 5 program committee members who again often asked competent referees for assistance. The final selection meeting of the program committee took place in Torino, on February 6,7 1989. The Program Committee (selection meeting participants underlined) consisted of:

G.Ausiello, Rome (co-chairman)

P.L.Curien, Paris

D.Dolev, Jerusalem/San Jose

P.H.Flajolet, Le Chesnay

M.Hennessy, Brighton

P.Mosses, Aarhus

F.P.Preparata, Urbana

U.Schöningh, Koblenz

A.Tarlecki, Warsaw

J.L.Balcazar, Barcelona

M.Dezani-Ciancaglini, Turin (co-chairman)

H.Ehrig, Berlin

Y.Gurevich, Ann Arbor

J.W.Klop, Amsterdam

J.Paredaens, Antwerp

A.Salomaa, Turku

P.Spirakis, Patras

E.Welzl, Graz.

We would like to express our gratitude to all the members of the Program Committee, and in particular to those who did the final selection, and to all the referees who assisted them (they are

listed on page V). Moreover we would like to thank very much all those who submitted papers to this ICALP Conference.

The program of ICALP 89 offered six invited talks which are included in this volume. The Program Committee thanks the invited lecturers

C. Böhm (Univ. di Roma "La Sapienza")

H. Ehrig (Univ. Berlin)

Z. Galil (Columbia Univ./Tel Aviv Univ.)

S. Micali (M.I.T.)

D. Perrin (Univ. Paris VII)

J.C. Reynolds (C.M.U.)

for having accepted the invitation.

Moreover we would like to thank the members of the Organizing Committee, which consisted of: F.Barbanera, P.Giannini, E.Giovannetti, F.Honsell, G.Lolli, S.Ronchi Della Rocca (chairman), B.Venneri.

We gratefully acknowledge all the institutions and corporations which supported this conference:

Apollo Computer SpA, Banca Popolare di Intra , Centro Ricerche FIAT, CNR (Comitato Scienze Matematiche, Comitato Scienze e Tecnologia dell'Informazione, Comitato Scienze di Ingegneria e Architettura), CSELT SpA, Delphi SpA, DGXIII-ESPRIT-Basic Research Actions, Dipartimento di Informatica dell'Università di Torino, Enidata SpA, FIAT SpA, Hewlett Packard Italiana SpA, IBM Italia, Istituto Bancario San Paolo di Torino, Regione Piemonte-Assessorato alla Cultura, Springer-Verlag, Tecciel SpA, Unione Industriale di Torino, Università di Roma "La Sapienza", Università di Torino.

Lastly we would like to mention the excellent cooperation with Springer-Verlag.

Torino, April 1989

G. Ausiello
M. Dezani-Ciancaglini
S. Ronchi Della Rocca

List of referees for ICALP 89

Aceto L.	Duponcheel L.	Konikowska B.
Ait-Kaci H.	Ehrhard T.	Konrad
Alevizos P.	Ehrich H.D.	Koymans K.
Alt H.	Engelfriet J.	Kreowski H.-J.
Alvarez C.	Enjalbert P.	Kreowski H.K.
Ambos-Spies K.	Ericson L.W.	Krivine J.L.
Apostolico A.	Fages F.	Kubiak R.
Arnold A.	Ferrari G.L.	Lackner G.
Astesiano E.	Floreen P.	Larsen K.
de Baer D.	Formann M.	Latteux M.
Baeten J.C.M.	Friboure L.	Lautemann C.
Balbo G.	Fuchs U.	Le Chenadec P.
Beauquier D.	Gabarro J.	Lee D.T.
Bednarczyk M.A.	Gacs P.	van Leeuwen J.
Behrends B.	Gambosi G.	Leszczyłowski J.
Berry G.	Ganzinger H.	Lozano A.
Bert M.	Gaudel M.-C.	Luccio F.
Bertling H.	Giannini P.	Luginbuhl D.
Bezem M.	Ginsburg S.	Mahr B.
Bilardi V.	Giolito P.	Mandrioli D.
Bini D.	Giovannetti E.	Mannila H.
Bondorf A.	van Glabbeek R.	Marchetti-Spaccamela A.
Bongiovanni G.	Glas R.	Martelli A.
Bonnier S.	Goeman H.	Mascari G.F.
Borzyszkowski A.	Gogolla	Mathieu C.
Boudol G.	Goldreich O.	Mauri G.
Brandenburg F.J.	Goldwurm M.	Mauw S.
Broy M.	Groote J-F.	Mayoh B.
Bruynooghe M.	Gyssens M.	Mazurkiewicz A.
Buntroch G.	Habel A.	Mehlhorn K.
Cardone F.	Haber F.	Meyer J.-J.Ch.
Ceri S.	Hagerup T.	Meyer auf der Heide F.
Casas R.	Hansen H.	Middelburg C.
Casley R.	Hanus M.	Middeldorp A.
Castellani I.	Hanus N.	Millington M.
Charron-Bost B.	Harju T.	Moggi E.
Chrząstowski-Wachtel P.	Harsu T.	Möhring R.H.
Cleaveland W.R.	Hauszler D.	Montanari U.
Coppo M.	Heering J.	Morain F.
Cosineau G.	Herich	Motwani R.
Costa G.	Hoene A.	Mulder H.
Dahlaus E.	Hoffmann B.	Mundhenk M.
Darondeau P.	Holenderski L.	Mundici D.
De Baer D.	Honkala J.	Nanni U.
Degano P.	Honsell F.	Naor M.
Delporte-Gallet C.	Hoogenboom H.J.	Naor S.
De Luca G.	Iwanowski S.	Nielsen M.
Dembinski P.	Janssens D.	Nielson F.
Deminet J.	Jantke K.-P.	Niemi V.
De Nicola R.	Jenner B.	Niwinski D.
Denttonen M.	Jomssus D.	Nurmi O.
Deransart P.	Jones N.D.	Olderog R.
De Schreye D.	Kanellakis P.	Olivière D.J.
Diaz J.	Karhumaki J.	Olivière H.
Di Battista G.	Kaul M.	Ottmann T.
Diks K.	Kirousis L.M.	Orponen P.
Dimitrovici C.	Kirsig B.	Pachocki W.
Drabent W.	Köbler J.	Pacholczyk J.W.

Padawitz P.	Sarrafzadeh M.	Vaandrager F.
Pantziou G.	Schinkel B.	Vanneschi M.
Parigot M.	Schmidt E.M.	Vardi M.
Paris D.	Schuler R.	Venneri B.
Partsch H.A.	Schwantrbach M.	de Vink E.
Pawlowski W.	Schwarzkopf O.	Vollmer H.
Paz A.	de Simone R.	Vrancken J.
Penczek W.	Skyum E.	de Vries F.-J.
Pepper P.	Smolensky R.	de Vrijer R.C.
Phillips I.	Smolka G.	Wagner H.
Pin J.E.	Soisalon-Soininen E.	Wagner K.
Pitts A.M.	Soria M.	Walker D.
Protasi M.	Steinby M.	Warmuth M.
Puel L.	Steyaert J.M.	Wegener H.
Reggio G.	Stirling C.	Wegener I.
Regnier M.	Stockmeyer L.	Weijland P.
Reischuk R.	Stomp F.	Wieringa R.
Rycko M.	Szalas A.	Winkowski J.
Rodenburg P.	Talamo M.	Winkel G.
Rodriguez-Artalejo M.	Tamassia R.	Wøginger G.
Ronchi Della Rocca S.	Tampakas B.	Wu M.M.
Rosenstiehl P.	Thierauf T.	Zaroliagis C.
Rote G.	Thomas W.	Zielonka W.
Rozenberg G.	Thomsen B.	Zucca E.
Rudak L.	Tiwari P.	
Rudzinski R.	Tollis I.G.	
Ruohonen K.	Torà J.	
Rycko M.	Torelli M.	
Sabadini N.	Ukkonen E.	
Santha M.	Urzyczyn P.	

TABLE OF CONTENTS

M. Abadi, L. Lamport, P. Wolper Realizable and Unrealizable Specifications of Reactive Systems	1
E. Allender Limitations of the Upward Separation Technique (Preliminary Version)	18
E. Allender, L.A. Hemachandra Lower Bounds for the Low Hierarchy (Extended Abstract)	31
R.A. Baeza-Yates, G.H. Gonnet Efficient Text Searching of Regular Expressions	46
D. Beauquier, J.E. Pin Factors of Words	63
P. Berman, J.A. Garay Asymptotically Optimal Distributed Consensus (Extended Abstract)	80
G. Bilardi, A. Moitra Time Lower Bounds for CREW-PRAM Computation of Monotone Functions	95
C. Böhm Subduing Self-Application (Invited Paper)	108
G. Brassard, C. Crépeau, M. Yung Everything in NP can be argued in <i>perfect</i> Zero-Knowledge in a <i>bounded</i> Number of Rounds	123
V. Breazu-Tannen, J. Gallier Polymorphic Rewriting Conserves Algebraic Strong Normalization and Confluence	137
V. Bruyere Completion of Finite Codes with Finite Deciphering Delay	151

F. Cardone	
Relational Semantics for Recursive Types and Bounded Quantification	164
B. Chazelle, H. Edelsbrunner, L.J. Guibas, M. Sharir	
A Singly-Exponential Stratification Scheme for Real Semi-Algebraic Varieties and its Applications	179
L. Colson	
About Primitive Recursive Algorithms	194
B. Courcelle	
The Definability of Equational Graphs in Monadic Second-Order Logic	207
K. Culik II, T. Harju	
Dominoes and the Regularity of DNA Splicing Languages	222
P. Darondeau, P. Degano	
Causal Trees	234
N. Dershowitz, S. Kaplan, D.A. Plaisted	
Infinite Normal Forms (Preliminary Version)	249
H. Ehrig, P. Pepper, F. Orejas	
On Recent Trends in Algebraic Specification (Invited paper)	263
J. Engelfriet, H.J. Hoogeboom	
Automata with Storage on Infinite Words	289
D. Eppstein, Z. Galil	
Parallel Algorithmic Techniques for Combinatorial Computation (Invited paper)	304
D. Feldman, R. Impagliazzo, M. Naor, N. Nisan, S. Rudich, A. Shamir	
On Dice and Coins: Models of Computation for Random Generation	319
P. Feldman, S. Micali	
An Optimal Probabilistic Algorithm For Synchronous Byzantine Agreement (Invited paper)	341

D. Fussel, V. Ramachandran, R. Thurimella	
Finding Triconnected Components by Local Replacements (Extended Abstract)	379
Z. Galil, K. Park	
An Improved Algorithm for Approximate String Matching	394
G. Gambosi, E. Nardelli, M. Talamo	
A Pointer-Free Data Structure for Merging Heaps and Min-Max Heaps	405
J.F. Groote, F. Vaandrager	
Structured Operational Semantics and Bisimulation as a Congruence (Extended Abstract)	423
T. Hagerup, M. Nowak	
Parallel Retrieval of Scattered Information	439
J. Håstad	
Tensor Rank is NP-complete	451
D.S. Hochbaum, J.G. Shanthikumar	
The Complexity of Nonlinear Separable Optimization	461
C.M. Kenyon-Mathieu, J.S. Vitter	
General Methods for the Analysis of the Maximal Size of Dynamic Data Structures (Extended Abstract)	473
M. Li, P.M.B. Vitanyi	
How to Share Concurrent Asynchronous Wait-Free Variables (Preliminary Version)	488
M. Li, P.M.B. Vitanyi	
A New Approach to Formal Language Theory by Kolmogorov Complexity (Preliminary Version)	506
G. Louchard, B. Randrianarimanana, R. Schott	
Dynamic Algorithms in D.E. Knuth's Model : A Probabilistic Analysis	521
Z. Manna, A. Pnueli	
Completing the Temporal Picture	534

Y. Mansour, B. Schieber, P. Tiwari	
Lower Bounds for Computations with the Floor Operation	559
I. Mason, C. Talcott	
Programming, Transforming and Proving with Function Abstractions and Memories	574
P. McKenzie, D. Thérien	
Automata Theory Meets Circuit Complexity (Extended Abstract)	589
K. Mehlhorn, W.J. Paul	
Two Versus One Index Register and Modifiable Versus Non-Modifiable Programs	603
C.H. Papadimitriou, M. Yannakakis	
Shortest Paths Without a Map (Extended Abstract)	610
F. Parisi-Presicce	
Modular System Design Applying Graph Grammars Techniques	621
D. Perrin	
Partial Commutations (Invited paper)	637
A. Pnueli, R. Rosner	
On the Synthesis of an Asynchronous Reactive Module	652
K. Pruhs, U. Manber	
The Complexity of Controlled Selection	672
P. Raghavan, M. Snir	
Memory Versus Randomization in On-Line Algorithms (Extended Abstract)	687
J.C. Reynolds	
Syntactic Control of Interference, Part 2 (Invited paper)	704
B. Steffen	
Characteristic Formulae	723
J. Toran	
A Combinatorial Technique for Separating Counting Complexity Classes	733

J.V. Tucker, J.I. Zucker

Horn Programs and Semicomputable Relations on Abstract Structures 745

G. Winskel

A Note on Model Checking the Modal ν -Calculus 761

G.Q. Zhang

DI-Domains as Information Systems (Extended Abstract) 773